

Simulation Temporal Data Module (STDM)

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Architecture Ecosystem: Structured Reality Standards™

Architecture Family: SIMULOS® — Simulation Governance Architecture

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Governed Space: Simulation Temporal Data

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Canonical Language: English (UCL™)

Governed Space

Simulation Temporal Data

Canonical Definition

Simulation Temporal Data Module (STDM) defines the governance framework through which the temporal properties of Simulation Inputs are identified, aligned, bounded, assessed, and preserved so that information entering a simulation remains temporally appropriate for the reality period, sequence, frequency, duration, synchronization, and dynamic behavior the simulation is intended to represent.

STDM establishes the temporal-input layer of SIDS by governing whether data is not merely available and technically usable, but temporally fit for the simulation context in which it is relied upon.

It governs the transition:

Simulation-Usable Input → Temporal Context → Temporal Alignment → Temporal Fitness Assessment → Temporal Conditions & Limitations → Temporally Governed Simulation Input

Operational Role

STDM serves as the fourth internal governance space of the Simulation Input & Data Standard (SIDS).

The preceding modules establish:

SIIM — What information is entering the simulation?

SDRM — What reality can that information legitimately represent?

SDTM — What happened to that information before simulation use?

STDM asks:

Does the information represent the correct time, temporal sequence, frequency, duration, synchronization, and dynamic context required by the simulation?

The governed progression is:

Simulation Input → Temporal Identity → Temporal Reference → Temporal Alignment
→ Temporal Coverage → Temporal Gap & Distortion Assessment → Temporal Fitness
Status → Simulation Use

Module Operational Space

STDM governs: temporal input identification, observation time, collection time, event time, processing time, effective time, simulation reference time, temporal provenance, temporal currency, temporal applicability, temporal alignment, temporal synchronization, temporal sequence, temporal ordering, temporal resolution, sampling frequency, temporal granularity, temporal coverage, duration coverage, temporal gaps, temporal discontinuities, stale data, delayed data, asynchronous data, temporal interpolation, temporal extrapolation, time-window selection, historical-period selection, seasonal coverage, cyclic behavior, trend representation, regime change, temporal drift, temporal dependencies, latency, temporal distortion, temporal uncertainty, temporal limitations, temporal status, temporal reassessment, and temporal traceability.

Module Function

STDM exists to prevent: old data being treated as current without assessment, current data being assumed relevant to historical conditions, observations from different periods being combined as if simultaneous, temporal order being lost during preprocessing, sampling frequency hiding fast system behavior, short observation periods being generalized to long-term dynamics, seasonal data being generalized across the entire year, temporal gaps disappearing during interpolation, asynchronous sensors appearing synchronized, delayed information being treated as real-time information, historical relationships being assumed stable after regime change, future extrapolation being represented as observation, and simulations producing precise outputs from temporally inappropriate input foundations.

Temporal Identity

Every material temporal Simulation Input should preserve sufficient information to determine when the represented information applies.

Temporal identity may include: timestamp, observation period, collection period, effective period, duration, sequence position, sampling interval, temporal resolution, or another relevant temporal reference.

Temporal Reference

A Temporal Reference establishes the time basis against which an input is interpreted.

Examples include:

UTC

Local Time

Mission Time

System Runtime

Relative Simulation Time

Event Time

Historical Period

or another defined reference system.

Where multiple references are used, their relationship should remain explicit.

Observation Time

Observation Time identifies when the represented phenomenon actually occurred or was observed.

This may differ from:

Collection Time

Processing Time

Transmission Time

or:

Simulation Consumption Time.

These distinctions may materially affect dynamic simulations.

Event Time vs Processing Time

SIMULOS® establishes:

When information was processed is not necessarily when the represented event occurred.

For real-time and event-driven simulations, confusing these temporal properties may distort causal or sequential relationships.

Temporal Currency

Temporal Currency asks:

Is the input sufficiently current for the simulation inquiry?

Currency is context-dependent.

Data from ten years ago may be suitable for one simulation and materially obsolete for another.

No Universal Freshness Rule

SIMULOS® establishes:

Temporal fitness cannot be determined from data age alone.

The relevant question is whether material properties of the represented reality have changed since the data was produced.

Temporal Applicability

An input may be temporally applicable only to a defined period.

For example:

Applicable: 2024–2026 operating configuration

does not automatically establish:

Applicable: 2030 projected configuration.

Temporal applicability should therefore remain bounded.

Temporal Alignment

Temporal Alignment establishes whether multiple Simulation Inputs refer to sufficiently compatible time periods or temporal states.

Conceptually:

Input A Time Basis + Input B Time Basis + Simulation Time Basis → Temporal Alignment Assessment Cross-Source Temporal Alignment

Different sources may contain observations from different periods.

For example:

Traffic Data — 2025

Road Infrastructure Data — 2022

Vehicle Population Data — 2026

Combining them may be legitimate, but the temporal differences should be assessed where they can materially affect simulation behavior.

Temporal Synchronization

Synchronization governs whether inputs that interact dynamically are sufficiently aligned in time.

This is particularly important for: sensor simulations, robotics, autonomous systems, control systems, industrial systems, communication systems, financial simulations, and real-time digital twins.

Synchronization Tolerance

Perfect synchronization may not always be required.

A simulation may define an acceptable temporal tolerance.

For example:

Maximum Sensor Alignment Difference: ± 20 ms

The tolerance should reflect the dynamics of the simulated system rather than technical convenience alone.

Temporal Sequence

Some simulation behavior depends upon the order of events.

Conceptually:

$A \rightarrow B \rightarrow C$

may produce a materially different outcome from:

$B \rightarrow A \rightarrow C$

STDM therefore preserves sequence where sequence affects simulated causality or behavior.

No Sequence Neutrality Principle

SIMULOS® establishes:

Temporal order must not be treated as irrelevant where changing event order can materially change the simulated outcome.

Temporal Resolution

Temporal Resolution defines the smallest meaningful time interval represented by an input.

Examples include: milliseconds, seconds, minutes, hours, days, months, or years.

The required resolution depends upon the dynamics being simulated.

Resolution Fitness

A simulation of millisecond control behavior may not legitimately rely solely upon hourly averages.

Conversely, millisecond data may be unnecessary for a multi-decade infrastructure simulation.

STDM therefore asks:

Is temporal resolution sufficient for the simulation question?

Sampling Frequency

Sampling Frequency defines how often observations are captured.

For example:

100 Hz

means:

100 observations per second.

Sampling frequency may determine whether fast behavior is visible or lost.

Sampling Adequacy

Where relevant system dynamics occur faster than the available sampling interval, material behavior may remain unobserved.

Therefore:

Available Frequency ≠ Sufficient Frequency Temporal Granularity

Temporal granularity describes the level at which information is temporally grouped.

For example:

individual transactions

may be aggregated into:

hourly totals

or:

daily averages.

Granularity changes may hide important temporal patterns.

Temporal Coverage

Temporal Coverage describes the period represented by the available input.

Examples:

One Hour

Three Months

Five Years

Complete Operational Lifecycle

Coverage should be evaluated relative to the Simulation Purpose and Scope.

Duration Sufficiency

A dataset may contain high-frequency observations while covering only a very short period.

High sampling density does not automatically establish sufficient duration coverage.

SIMULOS® therefore distinguishes:

Observation Density

from:

Observation Duration.

Seasonal Coverage

Where seasonal variation matters, the input basis should sufficiently represent relevant seasonal conditions.

For example:

data collected only during summer may not sufficiently support simulation of winter energy demand.

Cyclic Behavior

Relevant temporal cycles may include: daily, weekly, monthly, seasonal, annual, maintenance, production, economic, biological, or operational cycles.

STDM determines whether required cycles are sufficiently represented.

Temporal Variation

Simulation behavior may depend upon changes over time rather than static values.

Relevant patterns may include: acceleration, degradation, growth, decay, oscillation, accumulation, recovery, fatigue, adaptation, or escalation.

Trend Representation

Historical data may contain a trend.

The simulation should determine whether that trend: remains relevant, has stabilized, has reversed, has structurally changed, or should not be extrapolated.

Regime Change

A Temporal Regime Change occurs where relationships represented by historical information materially change.

Examples include: new regulation, new technology, infrastructure redesign, market restructuring, environmental change, system upgrade, behavioral shift, or operational-policy change.

Pre-Change $\neq$ Post-Change

SIMULOS® establishes:

Data representative of one temporal regime must not automatically be treated as representative of a materially different regime.

Temporal Drift

Temporal Drift occurs where the relationship between the input basis and current or target reality gradually changes over time.

Drift may affect: distributions, behaviors, environmental conditions, system performance, populations, or dependencies.

Temporal Gap

A Temporal Gap exists where required time periods contain insufficient or missing information.

Examples include: missing sensor intervals, absent seasonal periods, incomplete historical records, missing transition periods, or unobserved operational phases.

Temporal Gap Materiality

Not every gap matters.

A temporal gap becomes material where it could affect: simulated dynamics,

scenario behavior, output interpretation, validation, or downstream reliance.

Temporal Discontinuity

A discontinuity occurs where a time series changes abruptly because of: missing data, sensor replacement, system change, measurement-method change, configuration change, or actual physical change.

The cause should remain distinguishable where material.

Stale Data

Stale Data is information whose temporal distance from the simulation context may materially weaken its applicability.

Staleness is not determined solely by age.

It depends upon the rate at which the represented reality changes.

Stability-Relative Currency

A stable physical constant may remain temporally applicable indefinitely.

A rapidly changing market variable may become stale within minutes.

Therefore:

Required Currency $\propto$ Rate of Material Change

This relationship is conceptual, not numerical.

Delayed Data

Data may describe a current event but arrive after a delay.

Relevant delay may include: sensing delay, transmission delay, processing delay, aggregation delay, publication delay, or ingestion delay.

Latency

Temporal Latency is the difference between a relevant event time and the time at which the information becomes available to the simulation.

Latency may materially affect simulations involving: control, reaction, coordination, autonomous decision-making, or real-time operation.

Latency Must Not Disappear

SIMULOS® establishes:

Information available after an event must not be represented as information that was available before the event where availability timing affects simulated behavior.

Asynchronous Data

Inputs may arrive or be observed at different temporal frequencies.

Asynchronous information may require: alignment, buffering, interpolation, resampling, or explicit asynchronous modeling.

SDTM governs the transformation.

STDM governs whether the resulting temporal relationship is suitable.

Temporal Interpolation

Interpolation may fill values between known temporal observations.

STDM governs the temporal consequence of using those values.

Interpolated time points should remain distinguishable from direct observations where material.

Temporal Extrapolation

Temporal extrapolation estimates values beyond the observed period.

Extrapolated values should not silently acquire the temporal status of observations.

Historical Input

Historical information may be used to represent: prior conditions, baseline behavior, long-term patterns, rare events, or model calibration.

Historical status should remain explicit.

Historical ≠ Current

SIMULOS® establishes:

Historical relevance must be demonstrated where historical information is used to represent current or future conditions.

Future-Oriented Simulation Inputs

Future simulations may depend upon: forecasts, projections, assumptions, synthetic trajectories, planned configurations, or predicted environmental conditions.

Such information should remain distinguishable from historical observations.

Observed Time vs Projected Time

The simulation record should preserve whether a temporal input is:

Observed

Estimated

Projected

Forecast

Assumed

or:

Synthetic

where the distinction is material.

Temporal Dependency

One input may depend upon the temporal state of another.

For example:

Battery Temperature at Time t

may depend upon:

Load during $t-1$ through $t-n$.

Static treatment may therefore fail to represent the actual dependency.

Lag Relationship

Some effects occur after a delay.

Examples include: thermal response, economic response, biological response, fatigue, infrastructure degradation, or control-system delay.

Where material, lag relationships should remain represented.

Lead Relationship

Some information may act as an advance indicator of another condition.

Where such relationships are used in simulation, their temporal basis should remain explicit.

Temporal Causality

STDM does not establish universal causal truth.

It ensures that temporal relationships relied upon by the simulation do not become detached from the sequence and timing assumptions upon which they depend.

Temporal Distortion

A Temporal Distortion occurs where processing or input construction materially changes the timing characteristics of the represented phenomenon.

Examples include: smoothing away rapid events, shifting timestamps, averaging across event boundaries, resampling beyond adequate resolution, or synchronizing sources through unsupported assumptions.

Transformation-Induced Temporal Distortion

SDTM may identify the transformation.

STDM evaluates its temporal consequence.

This separation preserves the boundary:

SDTM → What transformation occurred?

STDM → Is the resulting temporal representation still fit for simulation use?

Temporal Uncertainty

Temporal information may itself be uncertain.

Examples include: approximate event time, clock error, synchronization uncertainty, uncertain historical dating, variable latency, or estimated temporal alignment.

STDM identifies temporal uncertainty as a distinct contribution to the broader Simulation Input Uncertainty governed by SIUM.

Clock and Reference Uncertainty

Different systems may use: different clocks, different synchronization mechanisms, different time zones, or imperfect timestamp precision.

Where material, these differences should be governed.

Temporal Boundary

A simulation may define explicit temporal boundaries.

For example:

Simulation Period: 2026–2030

Input applicability outside or inside that period should be assessed accordingly.

Temporal Scope Expansion

If the simulation period expands materially, previously accepted data may require reassessment.

For example:

Original Scope: 2026

becoming:

Expanded Scope: 2026–2040

may materially change the required temporal basis.

Temporal Fitness

Temporal Fitness describes whether an input is sufficiently appropriate for the temporal conditions of its intended simulation use.

Possible statuses may include:

Temporally Fit

Conditionally Temporally Fit

Partially Temporally Fit

Temporally Unfit

or:

Temporal Fitness Undetermined Temporal Fitness Is Context-Bound

A dataset may be temporally fit for one simulation and unfit for another.

Therefore:

Temporal Fitness = Input Time Basis relative to Simulation Time Requirement
Temporal Condition

An input may remain usable under explicit temporal conditions.

For example:

Applicable only to system configuration before 1 January 2027.

or:

Suitable only where input latency remains below 50 ms.

Temporal Restriction

A temporal restriction narrows where or when the information may legitimately be used.

Restrictions should propagate downstream where they materially affect simulation interpretation.

Temporal Limitation

A Temporal Limitation may include: insufficient duration, stale information, missing seasonal coverage, uncertain synchronization, unresolved regime change, excessive latency, inadequate sampling frequency, temporal gaps, or projected rather than observed values.

Temporal Reassessment Trigger

Reassessment may be required when: the Simulation Scope changes, the target time period changes, system dynamics change, a new regime begins, new data becomes available, sampling requirements change, latency changes, synchronization changes, a new system version is introduced, or temporal assumptions are materially revised.

No Temporal Inheritance Principle

SIMULOS® establishes:

Temporal fitness must not be inherited automatically from a previous simulation when the relevant time context has materially changed.

No Recency-to-Fitness Principle

Newer data is not automatically better data if it represents the wrong temporal condition.

No Historical-to-Current Principle

Historical data does not automatically represent current reality.

No Current-to-Future Principle

Current data does not automatically represent future reality.

No Density-to-Duration Principle

A large number of observations collected over a short period does not establish long-duration temporal coverage.

No Synchronization-by-Assumption Principle

Inputs from different clocks, systems, or periods must not be treated as synchronized merely because the simulation can technically combine them.

No Interpolation-to-Observation Principle

A temporally interpolated value remains generated information, not a direct observation.

No Latency Erasure Principle

A simulation must not silently give an actor, model, or system information earlier than that information would legitimately have been available in the represented reality.

Temporal Information Leakage

A particularly important failure occurs when future information is inadvertently introduced into an earlier simulated decision point.

Conceptually:

Information from $t+1 \rightarrow$ used at decision t

when the represented actor could not have known it.

This creates Temporal Information Leakage.

No Future Knowledge Principle

SIMULOS® establishes:

A simulated actor must not receive future information unless the Simulation Purpose explicitly requires and identifies that condition.

This is especially important for: financial simulations, predictive systems, autonomous decision systems, policy simulations, operational decision models, and historical counterfactual analysis.

Temporal Leakage Control

Where decision-making is simulated, the information available at each decision point should remain bounded by what would legitimately have been available at that time.

Conceptually:

Decision at $t \leq$ Information Available at t

unless the scenario explicitly introduces another information condition.

Temporal Snapshot Integrity

Where a simulation reconstructs a historical state, inputs should reflect a coherent temporal snapshot or explicitly governed multi-period reconstruction.

A historical simulation should not silently combine:

2022 knowledge + 2024 infrastructure + 2026 policy

and represent the result as:

Reality in 2022.

Temporal Input Set Coherence

Individual inputs may each be temporally acceptable while the complete Input Set is temporally incoherent.

STDM therefore assesses both:

Input-Level Temporal Fitness

and:

Input-Set Temporal Coherence.

Temporal Coherence

A Simulation Input Set is temporally coherent where material inputs can legitimately coexist within the represented simulation time context.

Temporal Conflict

A Temporal Conflict exists where two or more material inputs imply incompatible temporal states.

For example:

Infrastructure Version A active until June

while another input assumes:

Infrastructure Version B active since March.

The conflict should be resolved, bounded, or preserved explicitly.

Temporal Fitness Propagation

Material temporal limitations should propagate downstream.

Conceptually:

Temporal Limitation → Scenario / Execution Constraint → Output Interpretation
Limitation → Reliance Limitation

Later SIMULOS® Parent Standards govern those downstream stages.

STDM ensures the original temporal limitation remains visible.

Point-in-Time Reconstruction

STDM should enable questions such as:

What data was available at this simulated moment?

Which system version did the data represent?

Were the sources synchronized?

Which values were observed and which were interpolated?

Did the simulation use information from the future?

Was the input still temporally applicable?

Temporal Traceability

STDM establishes:

Simulation Input

↓

Temporal Identity

↓

Temporal Reference

↓

Temporal Coverage

↓

Temporal Alignment & Synchronization

↓

Temporal Gaps / Drift / Regime Change

↓

Temporal Fitness

↓

Conditions / Restrictions

↓

Simulation Use

Minimum Implementation Framework

1. Establish Temporal Identity

Identify the material temporal properties of each Simulation Input. 2. Define Simulation Temporal Requirements

Determine the: relevant period, resolution, frequency, duration, sequence, and synchronization requirements.

3. Assess Temporal Applicability

Determine whether each input represents the required time context. 4. Assess Temporal Alignment

Evaluate whether material inputs refer to sufficiently compatible temporal states. 5. Assess Resolution and Sampling

Determine whether the available temporal resolution and sampling frequency can represent materially relevant dynamics. 6. Assess Duration and Coverage

Determine whether the observation period covers required: cycles, seasons, transitions, operating phases, and long-term behavior.

7. Identify Temporal Gaps and Distortions

Record: missing periods, discontinuities, stale data, latency, asynchronous inputs, interpolation, extrapolation, and transformation-induced temporal effects.

8. Assess Drift and Regime Change

Determine whether historical temporal relationships remain applicable to the simulation context. 9. Control Temporal Information Leakage

Ensure simulated actors and systems receive only information legitimately available at the represented decision time unless explicitly governed otherwise. 10. Establish Temporal Conditions and Restrictions

Bound input use where temporal fitness is incomplete but manageable. 11. Assign Temporal Fitness Status

Determine whether the input or Input Set is: Temporally Fit, Conditionally Temporally Fit, Partially Temporally Fit, Temporally Unfit, or Temporal Fitness Undetermined.

12. Preserve Temporal Traceability

Maintain:

Input → Time Basis → Alignment → Coverage → Limitation → Temporal Fitness → Simulation Use

Governance Outputs

STDM may produce: Simulation Temporal Requirement, Temporal Input Identity Record, Temporal Reference Record, Observation-Time Record, Event-Time Record, Effective-Time Record, Simulation Reference-Time Record, Temporal Applicability Record, Temporal Currency Assessment, Temporal Alignment Assessment, Cross-Source Temporal Alignment Record, Synchronization Assessment, Synchronization Tolerance, Temporal Sequence Record, Temporal Resolution Assessment, Sampling Frequency Assessment, Temporal Granularity Record, Temporal Coverage Assessment, Duration Sufficiency Assessment, Seasonal Coverage Record, Cyclic Behavior Coverage Record, Trend Applicability

Record, Regime Change Record, Temporal Drift Record, Temporal Gap Record, Temporal Discontinuity Record, Stale Data Record, Latency Record, Asynchronous Data Record, Temporal Interpolation Record, Temporal Extrapolation Record, Temporal Dependency Record, Lag Relationship Record, Temporal Distortion Record, Temporal Uncertainty Record, Temporal Information Leakage Record, Temporal Snapshot Record, Temporal Conflict Record, Temporal Input Set Coherence Assessment, Temporal Condition Record, Temporal Restriction Record, Temporal Limitation Record, Temporal Fitness Status, Temporal Reassessment Trigger, and Simulation Temporal Data Traceability Map.

Use Case 1 — Autonomous Driving Simulation

Scenario

An autonomous-driving simulation combines: camera data captured at 30 Hz, radar data captured at 20 Hz, vehicle telemetry captured at 100 Hz, and pedestrian-location information updated every second.

The system being simulated makes control decisions every 50 milliseconds.
Application

STDM evaluates: sampling adequacy, synchronization, latency, event-time alignment, interpolation, and what information would actually have been available at each decision point.

The simulation is prohibited from using a pedestrian position update before that update would have been available to the vehicle. Result

The Input Set receives a governed Temporal Fitness Status with explicit synchronization tolerances and latency conditions.

The simulation can reconstruct the information state available to the autonomous system at every material decision point. Use Case 2 — Energy Demand Simulation

Scenario

A simulation estimates future electricity demand using three years of historical consumption data.

The available dataset contains: strong summer coverage, incomplete winter observations, a major industrial-demand shift in the final year, and no data after a significant energy-pricing reform.

Application

STDM identifies: seasonal coverage weakness, a temporal regime change, incomplete historical applicability, and limitations on projecting historical relationships forward.

Result

The data may remain:

Conditionally Temporally Fit

for defined baseline analysis but insufficient for unrestricted future-demand representation without additional temporal evidence or explicitly governed assumptions.

Architectural Position

Simulation Temporal Data is the fourth internal governance space of the Simulation Input & Data Standard (SIDS).

The complete SIDS progression is:

Simulation Input Identification

↓

Simulation Data Representativeness

↓

Simulation Data Transformation

↓

Simulation Temporal Data

↓

Simulation Input Uncertainty

SIIM establishes:

What information is entering the simulation?

SDRM establishes:

What reality can it legitimately represent?

SDTM establishes:

What happened to it before simulation use?

STDM establishes:

Is it temporally appropriate for the reality and dynamics being simulated?

The next module, SIUM — Simulation Input Uncertainty Module, governs the uncertainty carried by the complete Simulation Input basis after identification, representativeness, transformation, and temporal conditions have been established.

Foundational Principle

Simulation data becomes temporally legitimate only when its time basis, sequence, resolution, coverage, synchronization, latency, applicability, and

limitations remain aligned with the temporal reality the simulation is intended to represent.

Canonical Closing Statement

Simulation Temporal Data Module establishes the temporal governance layer of the Simulation Input & Data Standard by governing when Simulation Inputs apply, how their temporal references and sequences relate, whether their resolution, sampling frequency, duration, synchronization, and coverage are sufficient, and how temporal gaps, latency, drift, regime change, interpolation, extrapolation, temporal conflict, and future-information leakage affect simulation use. Through explicit temporal identity, applicability, alignment, coherence, conditions, restrictions, fitness status, reassessment triggers, and complete temporal traceability, STDm prevents technically usable information from being treated as temporally legitimate when its timing, age, sequence, availability, or dynamic context cannot support the reality the simulation claims to represent.

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